

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

PHYSICS — SSC-I (9th Grade)

Syllabus: Unit 1 — Physical Quantities & Measurement | Unit 2 — Kinematics | Unit 3 — Dynamics-I | Unit 4 — Dynamics-II | Unit 5 — Pressure & Deformation in Solids | Unit 6 — Work & Energy (till 6.4) | Unit 9 — Nature of Science

Total Marks: 65

Time Allowed: 2 hours 30 minutes

Version: 1

General Instructions:

- This paper has **three sections**: A (MCQs), B (Short Questions), C (Long Questions).
- Section A** (12 marks): 12 MCQs. Mark the correct bubble. Time: 15 minutes.
- Section B** (33 marks): 11 short questions $\times$ 3 marks each. Answer *either* the main question *or* the OR alternative.
- Section C** (20 marks): 4 long questions $\times$ 5 marks each. Answer *either* the main question *or* the OR alternative.
- Use **$g = 10 \text{ m/s}^2$** in numerical problems unless stated otherwise. Scientific calculator allowed.

SECTION A — Multiple Choice Questions ($12 \times 1 = 12$ Marks) [15 min]

Q.1: Choose the correct answer and fill in the corresponding bubble.

#	Question	A	B	C	D	Answer
1	How many significant figures are there in the measurement 0.00650 s?	3	2	5	6	☐ A ☐ B ☐ C ☐ D
2	A vernier caliper has a main scale with smallest division of 1 mm and 20 divisions on its vernier scale. Its least count is:	0.01 mm	0.05 mm	0.1 mm	0.5 mm	☐ A ☐ B ☐ C ☐ D
3	An athlete runs once around a circular track of length 400 m and returns to her exact starting point. The magnitude of her displacement is:	400 m	200 m	Zero	Cannot be determined	☐ A ☐ B ☐ C ☐ D
4	A scooter increases its speed from 4 m/s to 22 m/s in 6 seconds. Its acceleration is:	18 m/s^2	4.33 m/s^2	26 m/s^2	3 m/s^2	☐ A ☐ B ☐ C ☐ D
5	A swimmer pushes water backward with her hands and the water pushes her forward with an equal and opposite force. This is an example of:	Newton's 1st law	Newton's 2nd law	Newton's 3rd law	Conservation of mass	☐ A ☐ B ☐ C ☐ D
6	A rickshaw of mass 300 kg moves with a velocity of 6 m/s. Its momentum is:	$50 \text{ kg}\cdot\text{m/s}$	$300 \text{ kg}\cdot\text{m/s}$	$1800 \text{ kg}\cdot\text{m/s}$	$3600 \text{ kg}\cdot\text{m/s}$	☐ A ☐ B ☐ C ☐ D
7	For a rigid body to be in rotational (second condition) equilibrium, which of the following must be zero?	Net force	Net torque	Net momentum	Net velocity	☐ A ☐ B ☐ C ☐ D
8	A stone of mass 0.2 kg tied to a string of length 0.5 m moves in a horizontal circle at 4 m/s. The centripetal force acting on it is:	6.4 N	1.6 N	3.2 N	12.8 N	☐ A ☐ B ☐ C ☐ D
9	A spring stretches by 8 cm when a force of 16 N is applied to it. The spring constant of the spring is:	0.5 N/m	2 N/m	128 N/m	200 N/m	☐ A ☐ B ☐ C ☐ D
10	At sea level, the average atmospheric pressure is closest to:	7.6 mmHg	76 mmHg	760 mmHg	7600 mmHg	☐ A ☐ B ☐ C ☐ D
11	A 3 kg object is lifted to a height of 8 m. Taking $g = 10 \text{ m/s}^2$, its gravitational potential energy is:	24 J	30 J	80 J	240 J	☐ A ☐ B ☐ C ☐ D
12	Which branch of physics studies the fundamental particles and forces that create matter and radiation?	Astrophysics	Particle physics	Acoustics	Geophysics	☐ A ☐ B ☐ C ☐ D

SECTION B — Short Answer Questions (11 × 3 = 33 Marks)

Q.2: Attempt **all** questions. Answer *either* the main question *or* the OR alternative.

Part	Question	Marks	OR	OR Question	Marks
(i)	Write the following measurements in scientific notation : (a) 0.00072 m (b) 8,900,000 m. Also express the value in part (a) using an appropriate SI prefix.	3	OR	Name the SI prefixes used for the powers 10^2 , 10^{-2} and 10^{12} . Also convert 8 km into metres using scientific notation.	3
(ii)	Define least count of a vernier caliper. Calculate the least count if the smallest division on the main scale is 1 mm and the vernier scale has 25 divisions.	3	OR	Round off the following numbers to three significant figures : (a) 45.268 (b) 0.005672 (c) 9,432.	3
(iii)	A hiker walks 6 km towards north and then 8 km towards east. Calculate (a) the total distance covered, (b) the magnitude of her displacement .	3	OR	Differentiate between average speed and instantaneous speed . A bus covers 210 km in 3 hours. Calculate its average speed.	3
(iv)	A motorbike's speed increases uniformly from 5 m/s to 25 m/s in 8 seconds. Calculate (a) its acceleration , (b) the distance covered during this time.	3	OR	A ball is dropped from rest and takes 4 seconds to hit the ground. Taking $g = 10 \text{ m/s}^2$, find (a) its final velocity, (b) the height from which it fell.	3
(v)	A net force of 18 N acts on a trolley of mass 4 kg. Calculate its acceleration . What acceleration would the same force produce in a trolley of mass 12 kg?	3	OR	Explain, using the concept of inertia , why it is dangerous to jump off a bus while it is still moving.	3
(vi)	A hockey ball of mass 0.2 kg moving at 15 m/s is stopped by a goalkeeper in 0.1 s. Calculate (a) the impulse , (b) the average force exerted by the goalkeeper.	3	OR	A girl of mass 40 kg at rest on frictionless ice throws a 4 kg ball horizontally at 5 m/s. Using the law of conservation of momentum , calculate her recoil velocity.	3
(vii)	A mechanic applies a force of 60 N at the end of a spanner of length 30 cm to loosen a nut. Calculate the torque produced. What force must be applied with a shorter spanner of length 15 cm to produce the same torque?	3	OR	Ayesha (weight 480 N) sits 0.25 m from the pivot of a seesaw. At what distance from the pivot should Bilal (weight 400 N) sit on the other side to balance the seesaw?	3
(viii)	A ball of mass 0.4 kg is whirled in a horizontal circle of radius 0.8 m at a speed of 5 m/s. Calculate the centripetal force acting on it.	3	OR	A satellite orbits Earth in a circular path of radius 7000 km, completing one revolution in 6000 s. Calculate its average orbital speed .	3
(ix)	A spring extends by 6 cm when a mass of 3 kg is hung from it. Taking $g = 10 \text{ m/s}^2$, calculate the spring constant of the spring.	3	OR	A box weighing 240 N rests on the ground with a base area of 0.3 m^2 . Calculate the pressure it exerts on the ground.	3
(x)	A 2 kg ball is thrown with a speed of 12 m/s. Calculate its kinetic energy . What would be its new kinetic energy if its speed is doubled?	3	OR	Explain, with an example, how gravitational potential energy is converted into kinetic energy as an object falls. State the law of conservation of energy.	3
(xi)	Explain, with an example, why Newton's laws of motion needed modification by Einstein's theory of relativity. What does this tell us about the nature of scientific theories?	3	OR	Define any three fields of physics from the following: Biophysics, Astrophysics, Acoustics, Nuclear Physics. Give one real-world example of each.	3

SECTION C — Long Questions (4 × 5 = 20 Marks)

Q.3: Attempt **all** questions. Answer *either* the main question *or* the OR alternative.

Q.	Question	Marks	OR	OR Question	Marks
3(i)	Describe the construction and working of a vernier caliper. A vernier caliper has a least count of 0.01 cm. If the main scale reading is 3.2 cm and the 6th division of the vernier scale coincides with a main scale division, calculate the total reading.	5	OR	What is a screw gauge ? Define its pitch and least count. A screw gauge has a pitch of 1 mm and 100 divisions on its circular scale. If the main scale reading is 4 mm and the 32nd division of the circular scale coincides with the datum line, calculate the diameter of the wire being measured.	5
3(ii)	State Newton's second law of motion and write its mathematical form in terms of momentum. A cricket ball of mass 0.15 kg moving at 25 m/s is hit by a bat and rebounds at 15 m/s in the opposite direction, in a contact time of 0.01 s. Calculate (a) the change in momentum of the ball, (b) the average force exerted by the bat.	5	OR	State Newton's three laws of motion, giving one daily-life example of each. A goods truck of mass 2000 kg moving at 15 m/s collides with a stationary car of mass 800 kg and they lock together. Using the law of conservation of momentum, calculate their common velocity after the collision.	5
3(iii)	State the two conditions of equilibrium . A uniform metre rule is suspended horizontally from a string at its 50 cm mark. A weight of 8 N is hung at the 20 cm mark. Find the position at which a weight of 5 N must be hung on the other side to balance the rule (ignore the weight of the rule).	5	OR	Define centripetal force and orbital motion . A satellite of mass 500 kg orbits Earth in a circular path of radius 8000 km with an orbital period of 7200 s. Calculate (a) its average orbital speed, (b) the centripetal force acting on it.	5
3(iv)	State Pascal's principle and describe how it is applied in a hydraulic lift. A hydraulic lift has a small piston of area 0.004 m^2 and a large piston of area 0.8 m^2 . Taking $g = 10 \text{ m/s}^2$, what force must be applied on the small piston to lift a car of mass 1200 kg?	5	OR	Show that the pressure exerted by a liquid column of height h and density ρ is given by $P = \rho gh$. A water tank is filled to a depth of 6 m. Calculate the pressure exerted by the water at the bottom of the tank. (Density of water = 1000 kg/m^3 , $g = 10 \text{ m/s}^2$)	5

Invigilator's Signature: _____ Candidate's Signature: _____

Chapters Covered: Units 1-6 (till 6.4) & 9 -- End of Question Paper --